MODULE 05 · DOMAIN 1 - SOLUTION DESIGN AND ARCHITECTURE

End-to-End Architecture and Decomposition

Input, processing, output and feedback: what surrounds the model, and how to split the work.

DOMAIN 1 of 7 EXAM WEIGHT 17% MODULE 5 of 38

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The model is one box in the diagram. Most of the architecture is **what happens before and after it**.

One request, end to end

01
Input
Channel, identity, request.

02
Pre-process
Authorise, clean, redact, retrieve.

03
Model stage
Prompt, tools, structured output.

04
Post-process
Validate, apply rules, route.

05
Act or persist
Write, send, or hand to a person.

06
Feedback
Outcomes become evaluation data.

ON THE EXAM

Every stage before the model shrinks risk. Every stage after it catches what the model got wrong.

Which layer owns each component?

COMPONENT	LAYER
OCR on scanned payslips	Pre-processing
Redact account numbers the task does not need	Pre-processing
Confirm the underwriter may open this application	Pre-processing
Classify the document type	Model stage
Extract income and outgoings into fields	Model stage
Check fields against the schema and totals	Post-processing
Write validated fields to the case system	Post-processing

ON THE EXAM

Authorisation and redaction come before the model sees anything. Validation comes before anything is written.

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Business rules: in the prompt or in code?

IN THE PROMPT

Hoped for

- May vary run to run
- Hard to unit test
- Every change needs re-evaluation
- Opaque to auditors

IN CODE

Guaranteed

- Same input, same output
- Tested and versioned
- Auditable line by line
- Claude can still explain the result

THE TIEBREAKER

If a rule can be written as an if-statement or a formula, it belongs in code. Claude explains it.

Structure at the boundaries

- **Structured outputs** constrain the response to your JSON schema, so the next component can parse it.
- **A schema fixes the shape, not the truth.** Still check values and business rules.
- **Citations** tie claims to source passages when grounding matters.
- **Currency check:** structured outputs are generally available; they cannot be combined with citations.

> ON THE EXAM
>
> Shape from the schema, truth from validation and sources. You need both.

Is anyone waiting for the answer?

SOMEONE IS WAITING

Synchronous

- Stream the response
- Lighter tier where it is enough
- Keep the context lean
- Cache the stable prefix

NOBODY IS WAITING

Asynchronous

- Queue the work
- Message Batches at 50% off
- Most finish within an hour
- Hard limit of 24 hours

THE TIEBREAKER

Batch where nobody waits. Never where somebody does.

Absorbing a spike

- **A queue between intake and model calls** turns a surge into a delay, not an outage.
- **Rate limits** apply per minute to requests, input tokens and output tokens.
- **On a 429**, retry after the time the response gives, with backoff.
- **Prioritise** interactive traffic over background work.
- **Cached input reads** do not count towards the input-token rate limit.

> ON THE EXAM
>
> A spike handled by a queue is a slow minute. A spike handled by nothing is an incident.

The feedback loop

01
Capture
Approvals, edits, rejections, complaints.

02
Label
Keep the input, the output and the correction.

03
Add to evals
Real failures become test cases.

04
Change one thing
Prompt, retrieval, model or rule.

05
Measure and release
Compare against the eval set.

ON THE EXAM

Corrections that are overwritten and thrown away are the most expensive data you never kept.

Four ways to decompose a problem

SPLIT BY	REACH FOR IT WHEN
Stage	One call is doing too many jobs at once
Domain	Tools, knowledge and rules differ by area
Dependency	Some parts must finish before others start
Risk	Some steps need a gate or a person

> ON THE EXAM
>
> Good decomposition gives each piece one job, one owner and one way to be tested.

Reviewing a peer's design: the checklist

- **Authorisation before retrieval**, not when rendering the answer.
- **Secrets outside prompts and logs;** sensitive fields redacted in logs.
- **Deterministic rules in code;** output validated before any action.
- **A person before irreversible, high-impact actions.**
- **Spikes buffered;** nobody-waiting work batched.
- **A feedback path** from outcomes into the evaluation set.

> ON THE EXAM
>
> Each line is a question the exam has asked in some form. Learn the list.

> Corvane is laying out Mereside's mortgage document pipeline: OCR, redaction, access checks, document classification, field extraction, schema and total checks, and writing to the case system.
>
> The model risk committee wants to see which steps run before Claude receives any document content.

Which two components belong before the model call? Choose two.

A Redacting account numbers the extraction task does not need

B Checking that extracted totals add up

C Confirming the requesting underwriter may access this application

D Writing validated fields to the case system

E Classifying each document's type

◆ PAUSE AND DECIDE

Which two components belong before the model call? Choose two.

A Redacting account numbers the extraction task does not need

Data the model never sees cannot leak. Pre-processing.

B Checking that extracted totals add up

Needs the model's output. Post-processing.

C Confirming the requesting underwriter may access this application

Authorise before anything is retrieved or sent. Pre-processing.

D Writing validated fields to the case system

The last step, after validation. Post-processing.

E Classifying each document's type

A judgement on messy input: Claude's job, in the model stage.

THE SIGNAL

Authorise and minimise before the model. Validate and persist after it.

> Tallyhawk's support agent tells customers whether they qualify for a loyalty discount. The rule is fixed: annual plans with more than 50 seats get 15% off renewal. It is written into the system prompt.
>
> Audits find the agent sometimes offers the discount to monthly customers, and sometimes refuses eligible ones.

What is the most reliable fix?

A. Add more worked examples of eligible and ineligible customers to the prompt

B. Compute eligibility in code behind a tool, and have Claude explain the result

C. Move the support agent to the most capable model tier

D. Ask Claude to double-check its reasoning before answering

◆ PAUSE AND DECIDE

What is the most reliable fix?

A Add more worked examples of eligible and ineligible customers to the prompt

Improves the odds; still a rule applied by hope.

B Compute eligibility in code behind a tool, and have Claude explain the result

A fixed rule becomes a guarantee; Claude does the explaining.

C Move the support agent to the most capable model tier

Costs more and still cannot guarantee a fixed rule.

D Ask Claude to double-check its reasoning before answering

Helps sometimes; it is still the model checking the model.

THE SIGNAL

A rule you could write as an if-statement, applied inconsistently: move it into code.

> Mereside wants every complaint from the past month, about 3,000 of them, re-summarised in a standard format for a management report. The summaries are needed by nine the next morning.
>
> Every request shares the same long set of instructions. The finance director wants the cost kept down.

Which design fits best?

A. Stream each summary through the interactive API as fast as possible

B. A multi-agent pipeline that summarises and cross-checks each complaint

C. Send every request at midnight to the fastest tier, in parallel

D. Submit a message batch with the shared instructions cached, and collect results in the morning

◆ PAUSE AND DECIDE

Which design fits best?

A Stream each summary through the interactive API as fast as possible

Pays for speed nobody needs.

B A multi-agent pipeline that summarises and cross-checks each complaint

Many times the tokens for a fixed summarising job.

C Send every request at midnight to the fastest tier, in parallel

Likely to hit rate limits, and still full price.

D Submit a message batch with the shared instructions cached, and collect results in the morning

Nobody waits: half price, plus caching on the shared prefix.

THE SIGNAL

Nobody waiting, hours to spare, a shared prefix: batch it and cache it.

> Quillon's patient-portal assistant sees sharp surges for about twenty minutes after each batch of appointment reminders is sent. During surges, some requests fail with rate-limit errors and patients see an error message.
>
> Outside surges, latency and quality are fine.

What should Corvane add?

A Raise the maximum output tokens so replies finish in fewer calls

B Move the assistant to the most capable model tier

C A queue in front of the model calls, retrying after the time each rate-limit response gives, with portal traffic prioritised

D Send portal messages through the Message Batches API during surges

◆ PAUSE AND DECIDE

What should Corvane add?

A Raise the maximum output tokens so replies finish in fewer calls

Does not reduce the number of requests in a surge.

B Move the assistant to the most capable model tier

Different limits and higher cost; the surge still overruns them.

C A queue in front of the model calls, retrying after the time each rate-limit response gives, with portal traffic prioritised

Turns failures into short delays, and honours the limits.

D Send portal messages through the Message Batches API during surges

Patients are waiting. Batches can take up to a day.

THE SIGNAL

Short, predictable surges causing rate-limit failures: queue, honour retry-after, prioritise.

> Mereside's underwriters correct at least one extracted field in about 8% of document packs. They type corrections straight into the case system, overwriting Claude's output.
>
> Nobody can say which fields go wrong most often, or whether a recent prompt change helped.

What should Corvane add to the design?

A. Capture each correction alongside the original output as labelled data, track error rates by field, and feed the cases into the evaluation set

B. Raise the effort setting on the extraction call

C. Ask underwriters to email the team whenever they find an error

D. Run a second extraction call and keep the result where both agree

◆ PAUSE AND DECIDE

What should Corvane add to the design?

A Capture each correction alongside the original output as labelled data, track error rates by field, and feed the cases into the evaluation set

Closes the loop: measurable errors and real test cases.

B Raise the effort setting on the extraction call

A guess. Without measurement, nobody will know if it helped.

C Ask underwriters to email the team whenever they find an error

Manual, patchy, and unusable as data.

D Run a second extraction call and keep the result where both agree

May cut some errors, but still learns nothing from corrections.

THE SIGNAL

Corrections overwritten and lost, and no way to measure change: build the feedback loop.

> A peer's design for Quillon's prior-authorisation letters: Claude drafts each letter from the record and the insurer's criteria, and the pipeline submits it directly to the insurer's portal. A submission cannot be withdrawn.
>
> The design logs every letter for a monthly audit.

What is the most important change before build?

A Use the most capable model tier to reduce drafting errors

B Validate each letter against its schema and require clinician approval before submission

C Add a second call that rewrites each letter for a more professional tone

D Move the monthly audit to a weekly one

◆ PAUSE AND DECIDE

What is the most important change before build?

A Use the most capable model tier to reduce drafting errors

Fewer errors is not zero, and nothing stops the ones left.

B Validate each letter against its schema and require clinician approval before submission

A check and a human before an irreversible external action.

C Add a second call that rewrites each letter for a more professional tone

Polishes the letter; still submits it unchecked.

D Move the monthly audit to a weekly one

Still after the fact. The submission is already final.

THE SIGNAL

An irreversible action with no check before it: add validation and a human gate.

In five lines

- **Authorise and redact before the model;** validate before anything is written or sent.
- **Rules you can write as code belong in code.** Claude explains them.
- **Nobody waiting? Batch it, and cache the shared prefix.** Someone waiting? Stream.
- **Queues absorb spikes;** honour retry-after and prioritise interactive traffic.
- **Close the loop:** corrections become labelled data and evaluation cases.

> COMING UP
>
> Next: business value pillars, and how to tell efficiency from productivity from transformation.